Sofia Ramos

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EDUCATION

Olin College of Engineering | Needham, MA
Bachelor of Science in Mechanical Engineering | Expected May 2029 | GPA: 4.0
Relevant coursework: Mechanics of Solids and Structures, Mechanical Prototyping, Principles of Integrated Engineering

ENGINEERING EXPERIENCE AND PROJECTS

Prosthetic Hands and Wrists | **High School Capstone** | Spring 2025

- Designed and built three prototypes demonstrating wrist rotation, flexion, and deviation, refining the designs through four iterations using 3D printing, milling, and turning
- Programmed Arduino-controlled motors for wrist movement and gripping; tested each prototype with custom acrylic fixtures, with all three completing their intended tasks

Formula SAE Cockpit Engineer | Olin Electric Motorsports | Fall 2025–Present

- Designed the head restraint and mounting tabs in SolidWorks for the team's 2027 competition car
- Evaluated the mounting tabs under a 900 N rearward load using hand calculations and von Mises stress FEA, achieving a factor of safety of 4
- Prepared an engineering drawing and waterjet cutting file for the mounting tabs; plasma cut the aluminum head restraint

Desk Companion | MIT Hackathon | Spring 2026 | 2nd Place, Healthcare Track

- Owned the mechanical design of a posture and focus desk companion built in a 48-hour hackathon
- Designed and iterated phone holders in Onshape and integrated the hardware with teammates' posture-detection software for a live demo

WORK EXPERIENCE

Laguna Blanca High School | Santa Barbara, CA

Assistant Counselor | Summers 2024–2026

- Supervised 20+ campers ages 4–13 each week and led hands-on science activities, including micropipetting and bacterial growth experiments

TECHNICAL SKILLS

- **Programming:** MATLAB, C++ (Arduino)
- **CAD and Analysis:** SolidWorks, Onshape, FEA, Hand Calculations
- **Design Documentation:** Engineering Drawings, Bill of Materials (BOM)
- **Fabrication:** 3D printing, mill, lathe, plasma cutting
- **Languages:** English (Native), Spanish (Advanced)